

Jingjing(Alice) ZHANG

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ABOUT

- M.S. student in Biostatistics at New York University with training in statistical modeling and machine learning.
- Experience analyzing public health and biomedical data through academic and consulting projects.
- Seeking internships in biostatistics, epidemiology, or clinical/health data analysis where I can apply statistical modeling and analytic methods to generate data-driven insights that support public health and clinical research.

EDUCATION

New York University, School of Global Public Health

New York, USA

- **MS in Biostatistics | GPA: 3.89/4.0**

Sep 2025 - Present

Relevant Coursework: Biostatistics for Public Health, Epidemiology, Statistical Inference, Regression I & II (Linear & Categorical Data Analysis), Causal Inference, Statistical Programming in R, Data Management & Statistical Computing (SAS), Machine Learning

Dalian University of Technology

Dalian, CN

- **BSc in Mathematics**

Sep 2021 - Jun 2025

University of Cambridge, Girton College

Cambridge, UK

- **Deep Learning Summer Program | First Class**

Jul 2023 - Aug 2023

PROJECTS

Healthcare Data Analysis: MIND Diet & Veggie Meter Study | R

New York University Biostatistics Consulting Lab

Oct 2025 - Present

- Cleaned and prepared observational health datasets for statistical analysis using R (tidyverse).
- Analyzed patterns and trends in dietary behavior across demographic groups using observational datasets.
- Applied regression models to evaluate associations between dietary patterns and objective vegetable intake indicators, estimating effect sizes and 95% confidence intervals.
- Produced presentation-ready visualizations to summarize insights for non-technical stakeholders.

Breast Cancer Diagnosis Classification (Machine Learning) | R

New York University School of Global Public Health

Feb 2026 - May 2026

- Built and evaluated five classification models (Logistic Regression, LDA, KNN, Decision Tree, Random Forest) to predict tumor malignancy using 30 FNA biopsy features ($n = 569$).
- Performed data preprocessing, feature standardization, and 5-fold cross-validation to tune hyperparameters (KNN $k=14$, Random Forest $mtry=3$); assessed performance using AUC, sensitivity, and specificity.
- Achieved AUC up to 0.9997 with Random Forest and 100% sensitivity on test data, minimizing false negatives in a clinical screening context.
- Identified key predictive features (concave points, concavity, perimeter variability) aligned with known clinical indicators.

Hand Function and Depression Among U.S. Adults: NHANES 2021-2023 (Epidemiologic Study) | Stata

New York University School of Global Public Health

Feb 2026 - May 2026

- Conducted a cross-sectional study using NHANES 2021-2023 data ($n = 1,546$) to examine the association between difficulty using hands/fingers and depression among U.S. adults, using PHQ-9 as both a dichotomous and continuous outcome.
- Performed univariable, bivariable, and multivariable logistic and linear regression analyses in Stata, adjusting for diabetes status, smoking status, and prescription medication use, and tested general health status as an effect modifier.
- Found that participants with hand difficulty had significantly higher odds of depression after adjusting for confounders (aOR = 1.98, 95% CI: 1.48-2.65) and higher PHQ-9 scores ($\beta = 1.55$, SE = 0.29, $p < 0.001$); the interaction with general health status was not statistically significant.
- Authored a full epidemiologic report (abstract, methods, results, discussion) and presented findings as a scientific poster.

Time-Series Data Analysis: COVID-19 Epidemiological Trends | R

New York University School of Global Public Health

Dec 2025 - Jan 2026

- Analyzed U.S. state-level COVID-19 daily case data to examine temporal outbreak patterns and identify peak infection periods using 7-day moving averages.
- Processed and organized multi-state COVID-19 datasets to support time-series analysis.
- Compared regional trends in post-peak recovery patterns to identify key drivers of variation across states.
- Visualized epidemic trends using ggplot2 and tidyverse and produced a reproducible analytical report in R Markdown.

Reinforcement Learning Modeling: Multi-Agent Experiments (MPE) | Python

Cambridge Deep Learning Summer Program

Jul 2023 - Aug 2023

- Built a multi-agent reinforcement learning environment in Python using PettingZoo (simple_adversary_v2) to simulate adversarial resource capture scenarios.
- Implemented actor-critic and multi-agent deep deterministic policy gradient algorithms in Python to enable agents to learn optimal strategies.
- Modified reward functions to model real-world adversarial dynamics, incorporating pursuit and evasion incentives.
- Trained MADDPG agents over 10,000 episodes across two experimental scenarios (cooperative communication and competitive pursuit) and evaluated policy performance over 20 held-out episodes, observing convergence in average reward as agents learned pursuit and evasion behaviors.

SKILLS

- Statistical Methods: regression analysis, time-series analysis, probability modeling
- Programming: R (tidyverse, ggplot2), Python, SQL, SAS
- Data Tools: Excel, PowerPoint, R Markdown, data visualization
- Machine Learning: tree-based models, neural networks, reinforcement, logistic Regression, random forest, KNN, LDA